

Scoring and Analysis of Likert Scale: Few Approaches

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Abstract

To improve upon summative scoring of Likert Scale, two alternate methods of scoring have been proposed in this paper. The proposed methods of scoring are based on weighted sums where weights are taken as empirical probabilities and enable us to find cardinal scores for individuals as well as items as expected values satisfying conditions of additivity and linearity. Comparison of the methods was undertaken with an empirical data. Proposed methods of scoring Likert Scale increased reliability of the questionnaire in terms of Cronbach's alpha, tended to introduce independency among the items, helped in better exploring and interpreting the factors. Summated Likert scores did not follow Normal distribution but each of the proposed method of scoring passed test of normality. Thus, the transformed scores offer platform for undertaking almost all type of analysis being done for continuous quantitative variable following Normal distribution.

Key words: Likert-scale, Cardinal scores, Quantitative analysis, Reliability.

1. Introduction

Likert (1932) proposed a summated scale for the assessment of survey respondent's attitudes. Likert scaling presumes the existence of an underlying (or latent or natural) continuous variable whose value characterizes the respondents' attitudes and opinions. Likert scales are quite popular and are widely used in different areas like psychology, sociology, health care, marketing, attitude, preference, customers' quality perceptions or expectations, and of subjective well-being in health care, etc.

Individual item in Likert scale usually has odd number of response categories say 5 or 7. Descriptive response alternatives could be Strongly approved, Approved, Undecided, Disapproved, and Strongly disapproved. The response categories are usually assigned numbers like 1, 2, 3, 4, and 5 or by -2, -1, 0, 1 and 2 or any linear transformation of such numbers. However assigning successive integer values to scale categories has also been criticized for not being realistic.

Major limitations of Likert type data are:

- a) Respondents may avoid using extreme response categories(Central tendency bias) or agree with statements as presented (Acquiescanal bias) or try to

portray themselves or their organization in a more favourable light (Social desirability bias)

- b) One cannot assume that respondents perceive all pairs of adjacent levels as equidistant. Mid-point or neutral point or zero point is a perception.
- c) There is no hard and fast rule to decide on number of response alternatives. Number of response categories need not be an odd number. Five- or seven-point formats appear to be the most prevalent.
- d) Distance between response alternatives is assumed to be equidistant. If response categories are a, b, c, d, e the equidistant property can be achieved by either of the following two approaches.

* $b - a = c - b$ viz. 1, 2, 3, 4 5 or 0.1, 0.2, 0.3, 0.4, 0.5

* $\frac{b}{a} = \frac{c}{b}$ viz. 1, 2, 4, 8, 16

But verification of such assumptions is usually not done through data. Assigning successive integer values to scale categories has also been criticized.

- e) Non- Additivity: Naming the response classes by successive numbers does not mean we can do all further mathematical operations to the data generated from administration of such a questionnaire. Consider the hypothetical example where 50% respondents endorsed the response category ‘Most agree’ and the rest 50% went for “Most disagree”. Then the average score is neutral, but clearly, they are at two opposite poles. Thus, addition of responses of Likert items may not be meaningful. If addition is not meaningful then further statistics like mean, SD, Correlation, analysis like regression, estimation, testing etc. may not be meaningful. Moreover, there could be various response distributions even for a fixed number of response categories where the distributions have equal mean but different SDs.
- f) Generated responses: A major problem with Likert items is the generated responses consisting of a count of responses in each category. The resulting analysis is inherently limited, primarily to a frequency table, typically with relative and cumulative relative frequencies. However, assumption is often made that the Likert item is interval in nature and means and other statistics are commonly computed. It could be argued that this assumption is incorrect. The data generated by an instrument based on Likert items simply cannot be subjected to the more robust, more powerful and more subtle analysis available with quantitative data. Use of an ordinal scale implies a statement of “greater than” or “less than” without stating how much greater or lesser
- g) Continuous or discrete: Scores of Likert Items sometimes are treated as if they are not discrete. Few researchers are of the opinion that Likert scale (from strongly agree to strongly disagree) is ordinal and have to deal with this scale as a discrete scale. This implies that the mean, variance and covariance etc. computed from responses to a Likert scale may not be meaningful.
- h) Reliability: Reliability assessment might use the correlation between the item score and the total score or use a test-retest procedure. In any event, the

items not correlated with the total would be discarded. However, when items tend to measure various dimensions of the underlying trait, item correlations may be poor and if this is observed during analysis stage, one may not like to discard an item.

- i) Ordinal or Interval: The level of scaling obtained from Likert procedure is rather difficult to determine. The scale is clearly at least ordinal. Response categories tend to be sequential but not linear. Count of each category of response represents an ordinal variable. In order to achieve an interval scale, the properties on the scale variable have to correspond to differences in the trait on the natural variable. In other words, distance between any pair of response categories must be same. But it seems unlikely that the categories formed by the misalignment of the five responses will all be equal, the interval scale assumption seems unlikely.
- j) Normality: Assumption of Normality is generally not observed from data generated from Likert Scale. As a result, statistical analysis in the parametric set up cannot be meaningfully undertaken from data set generated from Likert scale.

Thus, need is felt to have new methods of scoring Likert scale so that the new scores are cardinal and have one to one correspondence with Real number system or a subset of it and enable us to perform analysis undertaken with quantitative data in parametric set up.

2. Literature review

Wu (2007) considered example of an injury scale of five categories represented by none, minor, moderate, severe, and fatal, the degree of injury seriousness between severe and fatal is more significant than that between none and minor. Thus, successive integers to the scale categories may not reflect the realistic differences in injury seriousness between or among scale categories.

While Jacoby et al (1971) suggest only three response categories, Dillman, et al. (2009) recommend that four or five categories should be used and Fink (1995) suggested five to seven. Foddy (1994) concludes that a minimum of seven categories is required to ensure scale validity and reliability. Nine-point format were used by researchers like Lee & Soutar (2010) and even a 15-point format was used by Chaiken & Eagly (1983).

Stevens (1946) mentioned that in the strict sense, statistics involving mean and SD ought not to be used with ordinal scales since mean and SD are in error to the extent that the successive intervals on the scale are unequal in size or not-equidistant. However, Muraki (1990) observed that if the data fits the Polytomous Rasch Model and fulfill the strict formal axioms of the said model, it may be considered as a basis for obtaining interval level estimates of the continuum. Chien-Ho Wo (2007) observed that transformation of scale data based on Snell's (1964) scaling procedure does not do much to pass the normality test. From item response theory, it can be seen that even large ordinal scales can be radically non-linear.

3. Objective

The paper proposes two new methods of scoring Likert scale which result in cardinal scores with one to one correspondence with Real number system or a subset of it and also to discuss advantages of such scoring methods and to compare among the methods primarily through empirical exploration.

4. Formal description

Suppose there are n – respondents who answered each of the m -items of a Likert questionnaire where each item has k -numbers of response categories.

Let X_{ij} be a general element of the basic data matrix of order $n \times m$ where n -individuals are in rows and m -items are in columns. X_{ij} represents score of the i -th individual for the j -th item. Value of X_{ij} ranges between 1 to k i.e. 1 to 5 for a 5-point scale

Note $\sum_{i=1}^n X_{ij}$ = Sum of scores of all individuals for the j -th item (Item Score for the j -th item)

$\sum_{j=1}^m X_{ij}$ = Sum of scores of all the items for i -th individual i.e. total score of the i -th individual (Individual score)

$\sum \sum X_{ij}$ = Sum of scores of all the individuals on all the items i.e. total test score

In addition, one can have another matrix $((f_{ij}))$ of order $m \times k$ showing frequency of i -th item to j -th response category. A row total will indicate frequency of that item and will be equal to the sample size (n). Similarly, a column total will indicate total number of times that response category was chosen by all the respondents. Grand total will be equal to sample size $\times$ number of items (mn).

Multinomial model for Likert responses: A k -dimensional random variable X with components $X_1, X_2, \dots, X_k$ has multinomial distribution or X follows Multinomial with parameters $(n; p_1, p_2, \dots, p_k)$ if the probability density function is given by:

$$P\{X_1 = n_1, \dots, X_k = n_k\} = \frac{n!}{n_1!n_2!\dots n_k!} p_1^{n_1} p_2^{n_2} \dots p_k^{n_k}$$

where $p_i \geq 0$ and $p_1 + p_2 + \dots + p_k = 1$ and $n_1 + n_2 + \dots + n_k = n$

The multinomial distribution $\text{Mult}(n, p_1, p_2, \dots, p_k)$ is the joint distribution of the k -random variables X_i . It is therefore a multivariate, discrete distribution with mean and variance as follows:

$$E[X_i] = np_i \text{ and } \text{Var}(X_i) = np_i(1 - p_i)$$

Thus, a Likert item with k- response categories can be viewed as throwing a k-faced dice, i.e. each throw of the dice may result into 1 or 2 or 3 or 4 or 5 for a 5-point scale. So responses to a Likert item can be assumed to follow a multinomial distribution

Maximum likelihood estimate of a Multinomial distribution is $p_j = \frac{f_j}{n}$ for $j = 1, 2, \dots, k$. Thus, estimate of p_j is simply the observed relative frequency of outcome j i.e. empirical probability. Reproduction theorem says that sum of independent multinomial random vectors with identical vectors of choice probabilities follow a multinomial distribution:

5. Methodology: Three approaches were adopted viz.

Approach 1: with total score of an individual as sum of his/her score on each item i.e. usual summative Likert scores. Here individual score lies between m to mk (for k point scale) and is discrete in nature. Similarly item scores is also discrete.

Approach 2: assigning uniform weights i.e. w_1, w_2, w_3, w_4 , and w_5 to the response categories which will remain unchanged for all items. Here, $\sum_{i=1}^k w_i = 1$ where w_i is the empirical probability of the i-th response category[$i = 1, 2, \dots, k$] and is calculated as $w_j = \frac{\sum_{i=1}^m f_{ij}}{\sum \sum f_{ij}}$ where number of items are 1, 2, m. In other words, weight for the i-th response category is the ratio of total frequency of the category (over all items) and grand total of the Item – Response Categories frequency matrix (mn).

Here, each item follows a Multinomial distribution with parameters $(n, w_1, w_2, w_3, w_4, w_5)$. Mean and variance of the i-th response category = $n \cdot w_i$ and $n w_i (1 - w_i)$ respectively. Correlation between i-th and j-th response category is $-\sqrt{\frac{w_i w_j}{(1 - w_i)(1 - w_j)}}$.

Total score of i-th item is taken as $\sum_{j=1}^k w_j f_{ij}$ where f_{ij} denotes frequency of the j-th response category of the i-th item and w_j denotes weight of the j-th response category. Similarly, score of the j-th response category is $\sum_{i=1}^m w_j f_{ij}$. Score of the i-th individual for the j-th item S_{ij} is equal to w_t if he/she responded to the t-th response category of the j-th item ($t = 1, 2, 3, 4, 5$ for a five point scale). Thus, both individual scores and item scores are in terms of probabilities or expected values and hence each provides measurements of continuous variable.

Approach 3: with different weights to different response categories of different items, so that sum of weights for each item is equal to one. Here, weight to the i-

th item and j-th response category is defined as $w_{ij} = \frac{f_{ij}}{\sum_{j=1}^k f_{ij}}$ i.e. ratio of cell frequency and total frequency of the item. . Clearly sum of weights for each item is equal to one and $\sum \sum w_{ij} = m$. However, sum of weights for a response category of all items is different from one. Score of an individual in the i-th item will be $\sum_{j=1}^k w_{ij} X_{ij}$ and score of j-th response category item will be $\sum_{i=1}^m w_{ij} X_{ij}$.

Here, each item will follow multinomial distribution with different values of parameters. Item scores are in terms of expectations. However, both item scores and individual scores are continuous variables.

Alternatively, one can find different weights to different item – response category combinations so that sum of weights (probabilities) of all cells is equal to one by choosing

$$w_{ij} = \frac{f_{ij}}{\sum \sum f_{ij}} \quad \text{Clearly } \sum \sum w_{ij} = 1.$$

Here, transformed cells scores follow multinomial distribution with larger number of parameters. However, weights in this method are linearly related to weights proposed in Approach 3. In fact, weights in this approach are equal to weights from approach 3 divided by number of items.

Weighted scores are in terms of expected values and hence provide measurements of a continuous variable.

6. Calculation of weights

Data : A test consisting of five Likert type items each with five response alternatives was administered to 100 respondents where “Strongly agree ” was assigned 5 and “ Strongly disagree” was assigned 1.

Empirical calculation of weights for various approaches is shown using the Item – Response Categories frequency matrix derived from the data. Here, number of items (m) = 5 , each item had 5 response categories i.e $K = 5$ and sample size $n = 100$.

Table – 1
Calculation of weights: Approach – 2

Items	*RC - 1	*RC - 2	*RC - 3	*RC - 4	*RC- 5	TOTAL
1	19	32	35	11	3	100
2	7	33	34	19	7	100
3	14	17	36	27	6	100
4	10	14	38	30	8	100
5	4	31	37	20	8	100
TOTAL	54	127	180	107	32	500
Weights to response categories	0.108	0.254	0.36	0.214	0.064	1.00

- $RC - i$ denotes i -th Response category

Here $w_1 = \frac{54}{500}$, $w_2 = \frac{127}{500}$ and so on

Table – 2
Calculation of weights: Approach - 3

Items	*RC- 1	*RC-2	*RC-3	*RC-4	*RC- 5	Total
1	0.19	0.32	0.35	0.11	0.03	1.00
2	0.07	0.33	0.34	0.19	0.07	1.00
3	0.14	0.17	0.36	0.27	0.06	1.00
4	0.10	0.14	0.38	0.30	0.08	1.00
5	0.04	0.31	0.37	0.20	0.08	1.00

Here, weight for the first response category of Item 1 is f_{11} divided by total frequency of the item i.e. sample size. Thus, $w_{11} = \frac{19}{100}$ and so on

7. Observations

In approach 2 and 3:

- Weights are taken as probabilities.
- Weights are obtained from data considering the frequencies or probabilities of Item – Response categories without involving assumptions of continuous nature or linearity or normality for the observed variables or the underlying variable being measured.
- It was assumed that there is no item with zero discriminating value i.e. there is no item with equal frequency for each response category and there is no item where all individuals recoded their response to only one response category. The assumption is reasonable since items with zero discriminating values are excluded as per method of test constructions.
- Item scores and individual scores are obtained as expected values and hence provides measurement of continues variables satisfying conditions of linearity since

$$E(x + y) = E(x) + E(Y)$$

$$E(\alpha x) = \alpha E(x)$$

$$E(\alpha x + \beta y) = \alpha E(x) + \beta E(y)$$
- Ranking of individuals are invariant under linear transformation for each of approach 1, 2 and 3
- Distribution of item scores follows multinomial distribution due to reproduction property of multinomial random variables for approach 2 and 3. Sum of item scores will follow multinomial distribution if the items are independent
- For large sample size, individual scores and item score may tend to follow normal distribution under each method. Primaface, each of approach 2 and 3

is likely to tend to normality but the proposition need to be tested empirically.

- The metric data and linearity of weighted scores by any of the approach 2 and 3 enable us to generate data that is cardinal (quantitative) to permit calculation of all descriptive statistics and also undertake relevant estimation, testing of hypothesis, relevant analysis used in multivariate statistics.

8. Analysis

Following analysis was undertaken based on empirical data described above:

8.1 Mean, Variance and Reliability of the questionnaire (Cronbach alpha) for the three approaches are shown in Table – 3 below:

Table – 3
Mean Variance and Reliability for various Approaches

Description	Approach - 1	Approach - 2	Approach – 3
Test mean	14.36	3.658	3.8464
Test variance	5.9904	0.6357	0.6893
Mean of Item – 1	2.47	0.66484	0.6613
Mean of Item – 2	2.86	0.72744	0.7384
Mean of Item – 3	2.94	0.7406	0.7758
Mean of Item – 4	3.12	0.77472	0.8744
Mean of Item – 5	2.97	0.7582	0.7965
Variance of Item – 1	1.0291	0.1347	0.1085
Variance of Item – 2	1.0604	0.0998	0.0709
Variance of Item – 3	1.2364	0.1209	0.1608
Variance of Item – 4	1.1456	0.1084	0.1861
Variance of Item – 5	0.9891	0.0919	0.0815
Cronbach's alpha	0.110552	0.157417	0.147983

Observations

* Approach 2 and Approach 3 resulted in reduction of test average and test variance i.e. the weighted scores made the data more homoscedastic.

* Scores as weighted sum resulted in higher values of alpha. Value of Cronbach's Alpha was highest for Approach 2 followed by Approach 3.

8.2 Rank correlation of individual score: Spearman ρ between individual scores obtained by various approach are shown in Table – 4.

Table - 4
Rank Correlation Matrix (Spearman ρ)

	Approach 1	Approach 2	Approach 3
Approach 1	1.0	0.444	0.306
Approach 2		1.0	0.909
Approach 3			1.0

Observations

- All the rank correlations are found to be significant.
- Low value of rank correlation between Approach 1 and other approaches indicate that ranks of individuals are different for different approaches. However high value of Spearman ρ between Approach 2 and 3 indicates that individual ranks are almost unchanged. In other words, ranks of respondents as obtained from Approach 2 remained more or less same when their ranks are computed by Approach 3.

8.3 Item Correlation matrix for each approach are shown in Table 5, 6 and 7

Table – 5
Item Correlation Matrix for Approach - 1

Items	1	2	3	4	5
1	1.0	0.168	0.096	0.003	0.123
2		1.0	(-) 0.007	0.033	0.045
3			1.0	*(-)0.330	(-) 0.11
4				1.0	0.172
5					1.0

*: Significant at 1% level

Table – 6
Item Correlation Matrix for Approach - 2

Items	1	2	3	4	5
1	1.00	0.051	(-)0.007	(-)0.074	0.038
2		1.00	0.009	0.111	0.098
3			1.00	0.052	0.156
4				1.00	(-)0.044
5					1.00

Table – 7
Item Correlation Matrix for Approach - 3

Items	1	2	3	4	5
1	1.0	0.037	0.024	(-) 0.005	(-) 0.055
2		1.0	0.117	0.143	0.101
3			1.0	(-) 0.019	0.189
4				1.0	(-) 0.120
5					1.0

Observations

- * Weighted sum used in Approach 2 and Approach 3 resulted in changes of magnitudes and signs of item-correlations
- * Significant correlation between item 3 and 4 as obtained in Approach 1 became insignificant in Approach 2 and also in Approach 3.
- * No significant correlations were found in Approach 2 and Approach 3 which imply that the weighted scores made the items more or less independent

8.4 Test of Normality

Attempt was made to test whether total score of respondents follow Normal distribution using Anderson – Darling test for Normality. It is one of most powerful statistical test for detecting departures from Normality. The underlying null hypothesis is that the variable under consideration is normally distributed. A large p-value corresponding to the test statistic ($p > 0.05$) would indicate normality. The test statistic is

$$AD = -N - \frac{2i-1}{N} (\ln(F(Y_i)) + \ln(1 - F(Y_{N+1-i})))$$

Table – 8
Values of test statistic and associated p-values

	Value of test statistics	p – values	Remarks
Approach 1	1.1153	0.0061	H_0 is rejected
Approach 2	0.4796	0.483315	H_0 is accepted
Approach 3	0.6824	0.07256	H_0 is accepted

Observations

- * It can be inferred that scores of respondents did not follow Normal distribution for Approach 1. In other words, summated Likert scores did not follow Normal distribution. This highlights limitation of Likert data. As a result, a number of statistical analysis, testing and estimation procedures which presume normality of data cannot be performed with usual summated scores of Likert type data
- * Respondents scores followed Normal distribution for Approach 2 and 3 i.e. non-linear transformations resulted in the desired property of normality. Thus, the transformed scores offer platform for undertaking almost all type of analysis being done for continuous quantitative variable following Normal distribution.
- * Higher p- value at Approach 2 in comparison to Approach 3 indicates better normality in case of Approach 2.

Similar approach was adapted to test whether item scores follow normal distribution or not. The test results showed that item scores are not normally distributed under any of the above said three approaches

8.5 Correlation matrix for the approaches: Correlation of individual scores obtained by various approach are shown in Table – 9.

Table – 9
Correlations between a pair of approaches

	Approach 1	Approach 2	Approach 3
Approach 1	1.0	0.445	0.307
Approach 2		1.0	0.927
Approach 3			1.0

Observations

*Maximum correlation was found between Approach 2 and Approach 3.

8.6 Factor structures of the approaches

Factor Analysis with orthogonal vari-max rotation was undertaken with item correlation matrix under each approach. The results are as follows

Table – 10
Results of Factor Analysis

Factor	Eigen values	Percentage of Variance explained	Cumulative percentage of variance explained	Remarks
APPROACH - 1				
1	1.382	27.638	27.638	Two factors explaining 52.507% of variance
2	1.243	24.869	52.507	
3	0.958	19.154	71.661	
4	0.787	15.741	87.403	
5	0.63	12.59	100	
APPROACH - 2				
1	1.204	24.077	24.077	Three factors explaining 66.867% of variance
2	1.105	22.10	46.177	
3	1.035	20.69	66.867	
4	0.891	17.829	84.696	
5	0.765	15.304	100	
APPROACH - 3				
1	1.277	25.544	25.544	Three factors explaining 68.755% of variance
2	1.152	23.042	48.586	
3	1.008	20.169	68.755	
4	0.825	16.492	85.246	
5	0.738	14.754	100	

Observations

- * Approach 1 gives two factors, combined effect of which explains only 52.50% of variance
 - * Each of Approach 2 and Approach 3 gives three factors explaining cumulatively 66.87% to 68.76% of variance respectively
 - * The results appear to be in line with item correlation matrix under each approach where each correlation was found to be insignificant except one in Approach 1 and also high correlation observed between Approach 2 and Approach 3
- Thus, the non-linear transformations tended to introduce independency of items

9. Limitations

Application of the proposed scoring of Likert items and Scale should take into account the following facts:

- * The methods take no account of the experiment design behind the data.
- * The methods are not applicable for items with zero discriminating value.
- * Irregularities in data should be within tolerance.
- * Test of Normality may be undertaken before application of the proposed methods of Scoring since distribution of individual score obtained from Approach-2 or Approach-3 is yet to be established.

10. Conclusions

Weighted scores where weights are data driven and proportional to probabilities helps to find total score of a respondent and also total score of an item as expected values and enable us to perform usual analysis for a continuous quantitative variable. Computation of weights considered the frequencies or probabilities of Item – Response categories without involving assumptions of continuous nature or linearity or normality for the observed variables or the underlying variable being measured.

It was assumed that there is no item with zero discriminating value i.e. there is no item with equal frequency for each response category and there is no item where all individuals recoded their response to only one response category. The assumption is reasonable since items with zero discriminating values are excluded as per method of test constructions.

Scores in terms of expected values resulted in higher reliability of the questionnaire. Approach 2 registered highest value of Cronbach's alpha among the three approaches. Such scores tended to introduce independency among the items. The Approach 2 and 3 resulted in a situation where the five items were almost independent and accordingly gave higher number of independent factors.

Thus, scores as per the proposed methods helped in better exploring and interpreting the factors.

Usual Likert type score did not follow normal distribution. However, weighted scores as per Approach 2 and also for Approach 3 resulted in the desired property of normality and are suitable for use in methods of analysis requiring assumption of normality. Thus, the proposed scores offer platform for undertaking almost all type of analysis being done for continuous quantitative variable following Normal distribution. For example, individual scores obtained through Approach 2 or Approach 3 tended to satisfy assumptions of AVOVA, regression analysis, t-test for testing equality of means, F-test for testing equality of variances, Discriminant analysis, etc. Proposed methods of Scoring conform better to Normality.

Thus, scoring of Likert Scale as per Approach 2 or Approach 3 has many desirable properties and avoids some of the major limitations of usual summative scores. However, high correlation (over 0.9) was found between Approach – 2 and Approach – 3. This may imply possible use of Approach 2 instead of Approach 3 primarily because of easiness to computation. Thus, Scoring method as proposed in Approach -2 is recommended for Likert-type data for clear theoretical advantages and easiness in calculations with minimum processing hour.

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